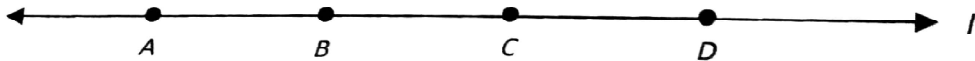


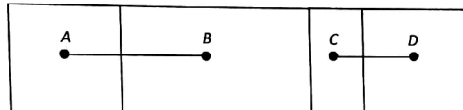
AMTI (NMTTC) - 2011
GAUSS CONTEST - PRIMARY LEVEL

- When a number n is divided by 10,000, the quotient is 1 and the remainder is 2011. The quotient and remainder when n is divided by 2011 are respectively.
 (A) 4, 1936 (B) 5, 1956 (C) 4, 0 (D) 5, 0
- Baskar is older than Prakash by one year minus one day. Baskar was born on January 1, 2005. The date of birth of Prakash is
 (A) January 2, 2006 (B) December 31st, 2005
 (C) December 31st 2004 (D) January 2nd 2004
- The points A, B, C and D are marked on a line l as shown in the figure.



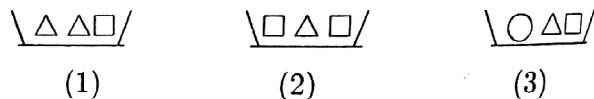
$AC = 12$ cm, $BD = 17$ cm, $AD = 22$ cm. Then $\left(\frac{AB}{CD}\right)$ is equal to

- (A) $\frac{5}{7}$ (B) $\frac{1}{2}$ (C) 2 (D) $\frac{7}{5}$
- The sum of all four digit numbers formed by using all the four digits of the number 2011 (including this number) is
 (A) 10877 (B) 12666 (C) 10888 (D) 12888
 - a, b, c are three natural numbers such that $a < b < c$ and $a + b + c = 6$. The value of c is
 (A) 1 (B) 2 (C) 3 (D) 1 or 2 or 3
 - A boy calculates the sum of the digits seen on a digital clock. (For eg., when the clock shows 20:20 then he sum is 4). The biggest digital sum that can be seen on a 24 hour clock is
 (A) 21 (B) 22 (C) 23 (D) 24
 - A thin rectangular strip of paper is 2011 cms long. It is divided into four rectangular strips of different sizes as in the figure.



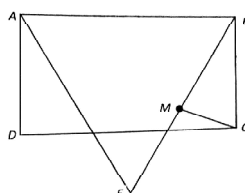
A, B, C, D are the centres of the rectangles (1), (2), (3) and (4) respectively. Then $(AB + CD)$ is equal to

- (A) $\frac{2011}{3}$ cms (B) $\frac{2011}{2}$ cms (C) $\frac{2011}{4}$ cms (D) $\frac{2(2011)}{3}$ cms
- Three trays have been arranged according to their weights in increasing order as follows :



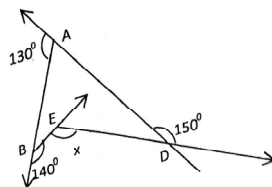
Where the symbols $\triangle = \square = \bigcirc$ are the three digits of numbers showing each of the weights. The position of the tray $\triangle = \square = \bigcirc$ lies

- (A) between (1) and (2) (B) between (2) and (3)
 (C) before tray (1) (D) after tray (3)
- ABCD is a rectangle in which $AB = 20$ cm, $BC = 10$ cm. An equilateral triangle ABE is drawn here and M is the midpoint of BE. Then $\angle CMB$ is equal to

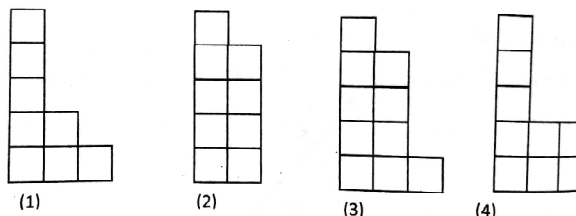


- (A) 70° (B) 75° (C) 65° (D) 90°

10. In the adjoining figure the value of x is

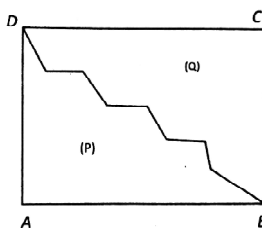


- (A) 110° (B) 130° (C) 120° (D) 125°
11. In the figure below pieces of squared sheets are shown. Each small square is 1 square unit.

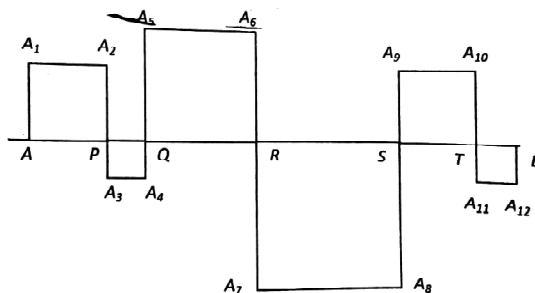


Two of them can be joined together without overlapping to form a rectangle. The area of this rectangle in square units is

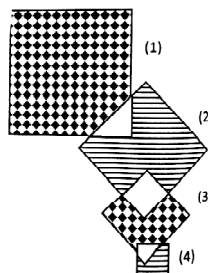
- (A) 18 (B) 19 (C) 16 (D) 17
12. ABCD is a rectangle and is divided into two regions P and Q by the broken Zig Zag line as shown. Then



- (A) The perimeter of the region P is greater than the perimeter of the region Q.
 (B) Area of region P is equal to the area of the region Q.
 (C) Perimeter of region P is equal to the perimeter of region Q.
 (D) Area of the region P is greater than the area of the region Q.
13. AB is a line segment 2011 cm long. Squares are drawn as in the diagram. The length of the broken line segment $AA_1A_2A_3A_4A_5A_6A_7A_8A_9A_{10}A_{11}A_{12}B$ is



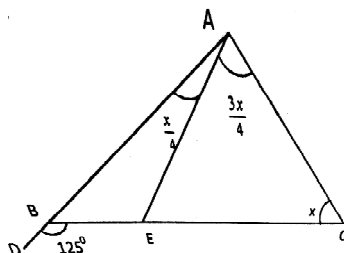
- (A) 2011 (B) 4022 (C) 6033 (D) 8044
14. In the adjoining figure there are four squares (1) with side 11 cm, (2) with side 9 cm (3) with side 7 cm and (4) with side 5 cm, (The area of the total dotted region) – (area of the lined region) is equal to (in cm^2)



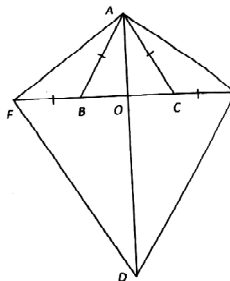
- (A) 56 (B) 64 (C) 78 (D) 89
15. The four quarters of the circle (or quadrants) of radius 1 cm, are rearranged to form the adjacent shape. The area of the second figure (in cm^2) is
- (A) 2 (B) 3 (C) 1 (D) 4

PART - B

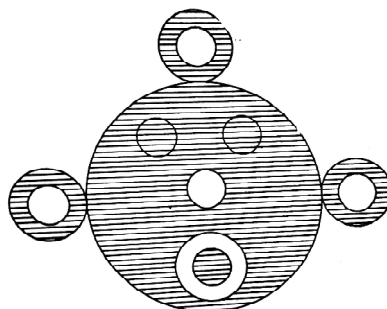
1. In the adjoining figure the value of x is _____.



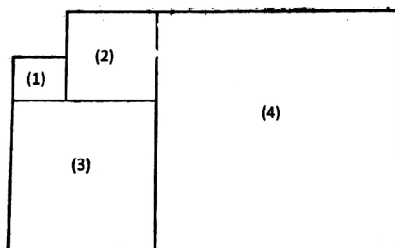
2. The maximum number of points of intersection of a circle and a triangle is m . The maximum number of points of intersection of two triangles is n . Then the value of $(m + n)$ is _____.
3. n is a two digit number. $P(n)$ is the product of the digits of n and $S(n)$ is the sum of the digits of n . If $n = P(n) + S(n)$ then the units digit of n is _____.
4. The number of two digit positive integers which have atleast one 6 as a digit is _____.
5. The sum of the digits of a two digit number is subtracted from the number. The units digit of the result is 6. The number of two digit numbers having this property is _____.
6. In the adjoining figure ABC and DEF are equilateral triangles $AB = BF = BC = CE = AC$. AD and EF cut at O and are perpendicular to each other. The number of right angled triangles formed in the figure is _____.



7. In the figure, the radius of each of the smallest circle is $\frac{1}{12}$ of the radius of the biggest circle. The radius of each of the middle sized circles is three times the radius of the smallest circle. The area of the shaded portion is _____ times the area of the biggest circle.



8. In the figure (1), (2), (3) and (4) are squares. The perimeter of the squares (1) and (2) are respectively 20 and 24 units. The area of the entire figure is _____.



9. The difference between the biggest and the smallest three digit numbers each of which has different digits is _____.
10. The degree measure of an angle whose complement is 25 % of its supplement is _____.
